Nonlinear functions — Part 3 answer key

#	ID	Answer	Difficulty
3.1	f54cf078-dbb9-4605-8b73-28108cb4c7b8	25	Hard
3.2	6e108e29-0eee-4507-8a4a-45c4b2c82383	5	Hard
3.3	397302ed-5a13-4b95-9bee-c626b3a9e221	D	Hard
3.4	b0fa9ba5-0ea2-43f1-b456-7834495ae933	D	Hard
3.5	230aed17-3fcc-4941-9290-a69a66f31d95	B	Hard
3.6	019546-DC	D	Hard
3.7	25d9ebe8-a480-4c88-9e5d-9b4803a4780c	C	Hard
3.8	0f7f7ce8-ffff-4ef3-be02-23e788d84376	B	Hard
3.9	08155-DC	B	Hard
3.10	a01bd482-116c-496d-b602-9a1cbb2d0dbd	-24	Hard
3.11	021312-DC	?	Hard
3.12	37d40ac0-74f9-4d4a-8168-1a7d28a874f2	C	Hard
3.13	de664bb9-acef-44d5-af54-80d0da2e25bb	D	Hard
3.14	b4b30a03-7260-43d7-a5d1-c60b998baa85	A	Hard
3.15	7a673830-c24f-450c-9dbf-0cbce334c119	B	Hard
3.16	917dab83-3019-497d-8e63-5d43c8147242	D	Hard
3.17	52a0019a-5b36-4d56-b8de-878dbc9fd5fd	-19	Hard
3.18	74ac8a77-8102-4458-aafe-4627ed4d7e68	B	Hard
3.19	06375-DC	?	Hard
3.20	019576-DC	A	Hard
3.21	cc127122-7a04-46bf-bc9e-396d6ef2a10a	A	Hard
3.22	63c46f21-221b-43b6-8f48-c9ab2ebf8eb3	-13/2, -6.5	Hard

3.23	2b6d8220-3a75-42df-a439-6f01d10147a0	3.44, 86/25	Hard
3.24	ad9d7217-a0fd-419c-b72d-02d82ef7e7c4	C	Hard
3.25	038001-DC	D	Hard
3.26	17ec22bd-b74b-4395-bda1-506081819636	D	Hard
3.27	e03b8f0c-00cb-46f3-af40-2ba9b200ab9c	A	Hard
3.28	025785-DC	D	Hard
3.29	65f2a8b0-dac1-4676-af53-3ed1d29659c3	609	Hard
3.30	08160-DC	D	Hard
3.31	5291f5d1-e675-4632-9051-76a6be583d50	D	Hard
3.32	e4d534f2-3a87-4887-9602-3feff6694dcb	B	Hard
3.33	7224c605-0224-443d-8b93-f567251f7a19	14	Hard
3.34	22bb3d70-ebff-46aa-8f1d-07a231d876e8	1728	Hard
3.35	e5403cac-ceb7-4643-b03d-c3f63fe73713	44	Hard
3.36	05720-DC	B	Hard
3.37	f78208f1-bb88-4092-9147-edda34ecda72	-12	Hard
3.38	ace58eff-0b88-4468-aeb9-2f5ff6974cd4	D	Hard
3.39	c52ea1c2-4bf1-4d4c-b86b-2dc95f66577b	336	Hard
3.40	2ae9a93c-e2a5-4e1e-9012-cdaf6592e09a	A	Hard
3.41	df267826-c54f-44df-bbc4-03bc6a5415ad	D	Hard
3.42	b47b1926-6e95-4552-8fc8-d6ec1af4c692	-2112	Hard
3.43	68f5f6ee-c704-4500-882b-ba9cb2d1a598	B	Hard
3.44	2812bef6-4e2d-4eae-b382-81985e9c9a12	A	Hard
3.45	bf8e1345-911b-42f4-ad9c-688a5ce90cb6	D	Hard
3.46	bb4ceae0-91ef-4a7e-8a48-9eecde3713f6	C	Hard
3.47	8defe594-1c91-42f1-b36d-c8fc7e515441	A	Hard
3.48	029595-DC	C	Hard

3.49	5b4efd5b-3a24-4e83-93ed-1065e3a90515	C	Hard
3.50	029650-DC	?	Hard
3.51	08364-DC	B	Hard
3.52	c061fa3e-bd9b-463c-94fa-bfb0819fdf17	A	Hard
3.53	158b94a1-9ef6-432b-9a5c-1435e1662fe7	A	Hard
3.54	27964a59-cacc-4c94-b077-538b9a79dbb5	48	Hard
3.55	cce2d985-f0ec-4127-81ef-a47c95e786ed	C	Hard
3.56	0fbeb13e-6775-487f-8f08-a6bf51a5bad6	D	Hard
3.57	e34a2287-9f25-422a-b964-03bc949e8c94	5	Hard
3.58	020462-DC	A	Hard
3.59	217d73a6-2b65-4411-bdbd-71f2d68adf48	D	Hard
3.60	5f865b4c-8ea2-42c5-aae5-8d455a024fd0	D	Hard
3.61	821bb037-c906-433b-a30b-6197a4337fb6	C	Hard
3.62	5d6ca18c-fd03-4a2f-89ab-d392b028d18b	20	Hard
3.63	019540-DC	D	Hard
3.64	845ca085-d722-4829-90e4-3d2962743b6a	A	Hard
3.65	a5817e09-28f5-4ad8-9f99-d3d5b3651286	B	Hard
3.66	f4d42535-f2e1-455c-b197-383e00a2a878	D	Hard
3.67	4fbe07d0-3558-4621-b4b3-5d8c574429ee	1.1, 11/10	Hard
3.68	e5580e8a-65b2-4cef-9984-856a4c58c355	B	Hard
3.69	c9709019-d048-4409-a506-d771fa061265	-4.9, -49/10	Hard
3.70	7bb5425b-4812-479b-b73a-acf842444820	B	Hard
3.71	10bfa82a-f80c-43fa-bf29-485d8932309c	A	Hard
3.72	c33f8edf-4790-4382-963c-36f09a033f73	D	Hard
3.73	212a6362-e804-4f52-add2-1e4b1d20094c	76	Hard
3.74	581cd95e-37cf-4835-9e46-c492ec9a8f24	B	Hard

3.75	406ed77a-b110-44a7-9d44-1c6ff1aa5918	A	Hard
3.76	29361323-e4f7-4672-a65d-d394fcdb43c0	5	Hard
3.77	1456688b-dc2f-4218-865c-e9bd32de6518	C	Hard
3.78	92038ced-e8e7-4472-a025-2fd98e61dd44	D	Hard
3.79	f8a81d21-4cde-4a6a-bab6-e67dcb97b10f	B	Hard
3.80	b2aba242-20bd-454b-aaca-55c42bdd0501	7	Hard
3.81	c99a75dd-0256-4d13-b211-eb05d63ef00d	A	Hard
3.82	e0e29be6-463f-477b-a2c1-4960db142fa1	A	Hard
3.83	6e51407b-255f-4375-91cf-b3145e516d19	110	Hard
3.84	24e16e9f-c530-4992-98b0-9848ddf76fbb	B	Hard
3.85	a09110a0-a8bc-49cf-ab9f-f198964cf20f	B	Hard
3.86	6233db14-f861-4237-b90c-a47a7b724601	35	Hard

Rationales

▼ 3.1 — 25

The correct answer is 25. The value of $g(7 - w)$ is the value of $g(x)$ when $x = 7 - w$, where w is a constant. Substituting $7 - w$ for x in the given equation yields $g(7 - w) = (7 - w)(7 - w - 2)(7 - w + 6)^2$, which is equivalent to $g(7 - w) = (7 - w)(5 - w)(13 - w)^2$. It's given that the value of $g(7 - w)$ is 0. Substituting 0 for $g(7 - w)$ in the equation $g(7 - w) = (7 - w)(5 - w)(13 - w)^2$ yields $0 = (7 - w)(5 - w)(13 - w)^2$. Since the product of the three factors on the right-hand side of this equation is equal to 0, at least one of these three factors must be equal to 0. Therefore, the possible values of w can be found by setting each factor equal to 0. Setting the first factor equal to 0 yields $7 - w = 0$. Adding w to both sides of this equation yields $7 = w$. Therefore, 7 is one possible value of w. Setting the second factor equal to 0 yields $5 - w = 0$. Adding w to both sides of this equation yields $5 = w$. Therefore, 5 is a second possible value of w. Setting the third factor equal to 0 yields $(13 - w)^2 = 0$. Taking the square root of both sides of this equation yields $13 - w = 0$. Adding w to both sides of this equation yields $13 = w$. Therefore, 13 is a third possible value of w. Adding the three possible values of w yields $7 + 5 + 13$, or 25. Therefore, the sum of all possible values of w is 25.

▼ 3.2 — 5

The correct answer is 5. Let x represent the width, in inches, of the rectangle. It's given that the length of the rectangle is 4 inches less than 7 times its width, or $7x - 4$ inches. The area of a rectangle is equal to its width multiplied by its length. Multiplying the width, x inches, by the length, $7x - 4$ inches, yields $x7x - 4$ square inches. It's given that the rectangle has an area of 155 square inches, so it follows that $x7x - 4 = 155$, or $7x^2 - 4x = 155$. Subtracting 155 from both sides of this equation yields $7x^2 - 4x - 155 = 0$. Factoring the left-hand side of this equation yields $7x + 31x - 5 = 0$. Applying the zero product property to this equation yields two solutions: $x = -\frac{31}{7}$ and $x = 5$. Since x is the rectangle's width, in inches, which must be positive, the value of x is 5. Therefore, the width of the rectangle, in inches, is 5.

▼ 3.3 — D

Choice D is correct. It's given that the mass of a sample of a substance decreases by 24% every 22.96 minutes. Therefore, after each 22.96-minute interval the mass is multiplied by $1 - 0.24$, or 0.76. It's also given that t represents the number of minutes after the lab analyst began observing the sample; therefore, $\frac{t}{22.96}$ represents the number of 22.96-minute intervals. The equation that could represent this model is $M = 100(0.76)^{\frac{t}{22.96}}$, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.4 — D

Choice D is correct. It's given that a and b are integer constants such that $a < b$ and $b < 0$. The y-coordinate of the y-intercept of the graph of a function is the value of the function when $x = 0$. Equation I is $px = a(4.1)^x + b$. Substituting 0 for x in this equation yields $p0 = a(4.1)^0 + b$, which can be rewritten as $p(0) = a(1) + b$, or $p(0) = a + b$. Therefore, the y-coordinate of the y-intercept of the graph of $y = p(x)$ is $a + b$. Equation I has a coefficient of a and a constant b. Thus, the value of the y-coordinate of the y-intercept, $a + b$, is not displayed as a constant or coefficient in equation I. Equation II is $rx = a(4.1)^{x+b}$. Similarly, substituting 0 for x in this equation yields $r0 = a(4.1)^{0+b}$, which can be rewritten as $r0 = a(4.1)^b$. Since $b < 0$, the value of $(4.1)^b$ is not equal to 1. Thus, the value of $a(4.1)^b$ is not equal to a. The value of $a(4.1)^b$ is also not equal to b, since there is no pair of integers a and b such that $a(4.1)^b = b$. Thus, the value of the y-coordinate of the y-intercept for this

equation is $a(4.1)^b$, which is not displayed in equation II as a constant or coefficient. The coefficient in equation II is a, and the only constant, b, appears in the exponent. Therefore, neither equation I nor equation II displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.5 — B

Choice B is correct. An x-intercept of a graph in the xy-plane is a point on the graph where the value of y is 0. Substituting 0 for y in the given equation yields $0 = 2x - dx + dx + gx - d$. By the zero product property, the solutions to this equation are $x = d$, $x = -d$, $x = -g$, and $x = d$. However, $x = d$ and $x = d$ are identical. It's given that d and g are unique positive constants. It follows that the equation $0 = 2x - dx + dx + gx - d$ has 3 unique solutions: $x = d$, $x = -d$, and $x = -g$. Thus, the graph of the given equation has 3 distinct x-intercepts.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 3.6 — D

Choice D is correct. The positive x-intercept of the graph of $y = h(x)$ is a point (x, y) for which $y = 0$. Since $y = h(x)$ models the height above the ground, in feet, of the projectile, a y-value of 0 must correspond to the height of the projectile when it is 0 feet above ground or, in other words, when the projectile is on the ground. Since x represents the time since the projectile was launched, it follows that the positive x-intercept, $(x, 0)$, represents the time at which the projectile hits the ground.

Choice A is incorrect and may result from misidentifying the y-intercept as a positive x-intercept. Choice B is incorrect and may result from misidentifying the y-value of the vertex of the graph of the function as an x-intercept. Choice C is incorrect and may result from misidentifying the x-value of the vertex of the graph of the function as an x-intercept.

▼ 3.7 — C

Choice C is correct. For the form of the function in choice C, $f(x) = 1281.6^{x-1}$, the value of $f(1)$ can be found as 1281.6^{1-1}, which is equivalent to 1281.6^{0}, or 128. Therefore, $k = 128$, which is shown in $f(x) = 1281.6^{x-1}$ as the coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.8 — B

Choice B is correct. It's given that $f(x) = x - 44x - 46$, which can be rewritten as $f(x) = x^2 - 90x + 2{,}024$. Since the coefficient of the x^2-term is positive, the graph of $y = f(x)$ in the xy-plane opens upward and reaches its minimum value at its vertex. For an equation in the form $f(x) = ax^2 + bx + c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 - 90x + 2{,}024$, $a = 1$, $b = -90$, and $c = 2{,}024$. It follows that the x-coordinate of the vertex is $-\frac{-90}{21}$, or 45. Therefore, $f(x)$ reaches its minimum when the value of x is 45.

Choice A is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = f(x)$ in the xy-plane.

Choice C is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = f(x)$ in the xy-plane.

Choice D is incorrect. This is the y-coordinate of the vertex of the graph of $y = f(x)$ in the xy-plane.

▼ 3.9 — B

Choice B is correct. The term (1 + r) represents a percent change. Since the population is decreasing, the percent change must be between 0% and 100%. When the percent change is expressed as a decimal rather than as a percent, the percentage change must be between 0 and 1. Because (1 + r) represents percent change, this can be expressed as $0 < 1 + r < 1$. Subtracting 1 from all three terms of this compound inequality results in $-1 < r < 0$.

Choice A is incorrect. If $r < -1$, then after 1 year, the population P would be a negative value, which is not possible. Choices C and D are incorrect. For any value of $r > 0$, $1 + r > 1$, and the exponential function models growth for positive values of the exponent. This contradicts the given information that the population is decreasing.

▼ 3.10 — -24

The correct answer is -24. Since the graph passes through the point $(0, -6)$, it follows that when the value of x is 0, the value of y is -6. Substituting 0 for x and -6 for y in the given equation yields $-6 = 2(0)^2 + b(0) + c$, or $-6 = c$. Therefore, the value of c is -6. Substituting -6 for c in the given equation yields $y = 2x^2 + bx - 6$. Since the graph passes through the point $(-1, -8)$, it follows that when the value of x is -1, the value of y is -8. Substituting -1 for x and -8 for y in the equation $y = 2x^2 + bx - 6$ yields $-8 = 2(-1)^2 + b(-1) - 6$, or $-8 = 2 - b - 6$, which is equivalent to $-8 = -4 - b$. Adding 4 to each side of this equation yields $-4 = -b$. Dividing each side of this equation by -1 yields $4 = b$. Since the value of b is 4 and the value of c is -6, it follows that the value of bc is $4(-6)$, or -24.

Alternate approach: The given equation represents a parabola in the xy-plane with a vertex at $(-1, -8)$. Therefore, the given equation, $y = 2x^2 + bx + c$, which is written in standard form, can be written in vertex form, $y = a(x - h)^2 + k$, where (h, k) is the vertex of the parabola and a is the value of the coefficient on the x^2 term when the equation is written in standard form. It follows that $a = 2$. Substituting 2 for a, -1 for h, and -8 for k in this equation yields $y = 2(x - (-1))^2 + (-8)$, or $y = 2(x + 1)^2 - 8$. Squaring the binomial on the right-hand side of this equation yields $y = 2(x^2 + 2x + 1) - 8$. Multiplying each term inside the parentheses on the right-hand side of this equation by 2 yields $y = 2x^2 + 4x + 2 - 8$, which is equivalent to $y = 2x^2 + 4x - 6$. From the given equation $y = 2x^2 + bx + c$, it follows that the value of b is 4 and the value of c is -6. Therefore, the value of bc is $4(-6)$, or -24.

▼ 3.11 — ?

The correct answer is 210. Since -5, 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x+5)(x-6)(x-7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.

▼ 3.12 — C

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k, which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24, of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant.

Choice D is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

▼ 3.13 — D

Choice D is correct. It's given that $fx = x - 14x + 19$, which can be rewritten as $fx = x^2 + 5x - 266$. Since the coefficient of the x^2-term is positive, the graph of $y = fx$ in the xy-plane opens upward and reaches its minimum value at its vertex. The x-coordinate of the vertex is the value of x such that fx reaches its minimum. For an equation in the form $fx = ax^2 + bx + c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $fx = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the x-coordinate of the vertex is $-\frac{5}{21}$, or $-\frac{5}{2}$. Therefore, fx reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x-value of the midpoint between the two x-intercepts of the parabola. Since it's given that $fx = x - 14x + 19$, it follows that the two x-intercepts of the graph of $y = fx$ in the xy-plane occur when $x = 14$ and $x = -19$, or at the points $14, 0$ and $-19, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2, is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $14, 0$ and $-19, 0$ is $\frac{14 + -19}{2}, \frac{0 + 0}{2}$, or $-\frac{5}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane.

Choice B is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = fx$ in the xy-plane.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.14 — A

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is 1 inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is 11 inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.15 — B

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point $0, \ 8$. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4. Each of the given choices is in the form $fx = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $fx = -3^x + k$ yields $fx = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.16 — D

Choice D is correct. The volume of a right rectangular prism can be represented by a function V that gives the volume of the prism, in cubic inches, in terms of the length of the prism's base. The volume of a right rectangular prism is equal to the area of its base times its height. It's given that the length of the prism's base is x inches, which is 7 inches more than

the width of the prism's base. This means that the width of the prism's base is $x - 7$ inches. It follows that the area of the prism's base, in square inches, is $xx - 7$ and the volume, in cubic inches, of the prism is $xx - 79$. Thus, the function V that gives the volume of this right rectangular prism, in cubic inches, in terms of the length of the prism's base, x, is $Vx = 9xx - 7$.

Choice A is incorrect. This function would give the volume of the prism if the height were 9 inches more than the length of its base and the width of the base were 7 inches more than its length.

Choice B is incorrect. This function would give the volume of the prism if the height were 9 inches more than the length of its base.

Choice C is incorrect. This function would give the volume of the prism if the width of the base were 7 inches more than its length, rather than the length of the base being 7 inches more than its width.

▼ 3.17 — -19

The correct answer is -19. It's given that function f is defined by $fx = a^x + b$, where a and b are constants and $a > 0$. It's also given that the graph of $y = fx$ in the xy-plane has a y-intercept at $0, -25$ and passes through the point $2, 23$. Since the graph has a y-intercept at $0, -25$, $f0 = -25$. Substituting 0 for x in the given equation yields $f0 = a^0 + b$, or $f0 = 1 + b$, and substituting -25 for $f0$ in this equation yields $-25 = 1 + b$. Subtracting 1 from each side of this equation yields $-26 = b$. Substituting -26 for b in the equation $fx = a^x + b$ yields $fx = a^x - 26$. Since the graph also passes through the point $2, 23$, $f2 = 23$. Substituting 2 for x in the equation $fx = a^x - 26$ yields $f2 = a^2 - 26$, and substituting 23 for $f2$ yields $23 = a^2 - 26$. Adding 26 to each side of this equation yields $49 = a^2$. Taking the square root of both sides of this equation yields $\pm 7 = a$. Since it's given that $a > 0$, the value of a is 7. It follows that the value of $a + b$ is $7 - 26$, or -19.

▼ 3.18 — B

Choice B is correct. It's given that function h is defined by $h(x) = a^x + b$, where a and b are positive constants. It's also given that the graph of $y = h(x)$ in the xy-plane passes through the point $(0, 10)$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$. Since $a^0 = 1$, this equation can be rewritten as $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $h(x) = a^x + b$ yields $h(x) = a^x + 9$. It's also given that the graph of $y = h(x)$ in the xy-plane passes through

the point $(2, 13)$. Substituting 2 for x and 13 for $h(x)$ in the equation $h(x) = a^x + 9$ yields $13 = a^2 + 9$. Subtracting 9 from both sides of this equation yields $4 = a^2$. Taking the square root of both sides of this equation yields $2 = a$ and $-2 = a$. Since a is positive, it follows that $2 = a$. Since $9 = b$ and $2 = a$, it follows that the value of ab is $(2)(9)$, or 18.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.19 — ?

The correct answer is 13. Let w represent the width of the rectangular garden, in feet. Since the length of the garden will be 5 feet longer than the width of the garden, the length of the garden will be $w+5$ feet. Thus the area of the garden will be $w(w+5)$. It is also given that the area of the garden will be 104 square feet. Therefore, $w(w+5)=104$, which is equivalent to $w^2+5w-104=0$. Factoring this equation results in $(w+13)(w-8)=0$. Therefore, $w=8$ and $w=-13$. Because width cannot be negative, the width of the garden must be 8 feet. This means the length of the garden must be $8+5=13$ feet.

▼ 3.20 — A

Choice A is correct. In 1 year, there are 4 quarter years, so the number of quarter years, q, is 4 times the number of years, t ; that is, $q=4t$. This is equivalent to $t=\frac{q}{4}$, and substituting this into the expression for M in terms of t gives $M=1{,}800(1.02)^{\frac{q}{4}}$.

Choices B and D are incorrect and may be the result of incorrectly using $t=4q$ instead of $q=4t$. (Choices B and D are nearly the same since 1.02^{4q} is equivalent to $(1.02^4)^q$, which is approximately 1.082^q.) Choice C is incorrect and may be the result of incorrectly using $t=4q$ and unnecessarily dividing 0.02 by 4.

▼ 3.21 — A

Choice A is correct. The *y*-intercept of a graph in the *xy*-plane is the point where $x = 0$. Substituting 0 for x in the given equation, $y = 576^{2x+2}$, yields $y = 576^{20+2}$, which is equivalent to $y = 576^2$, or $y = 331{,}776$. Therefore, the graph of the given equation in the *xy*-plane has a *y*-intercept of $0, 331{,}776$. It follows that $r = 0$ and $s = 331{,}776$. Thus, the equivalent equation $y = 331{,}776^{x+1}$ displays the value of s as the base.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.22 — -13/2, -6.5

The correct answer is $-\frac{13}{2}$. The value of x for which fx reaches its minimum can be found by rewriting the given equation in the form $fx = x - h^2 + k$, where fx reaches its minimum, k, when the value of x is h. The given equation, $fx = x - 2x + 15$, can be rewritten as $fx = x^2 + 13x - 30$. By completing the square, this equation can be rewritten as $fx = x^2 + 13x + \frac{13}{2}^2 - 30 - \frac{13}{2}^2$, which is equivalent to $fx = x + \frac{13}{2}^2 - \frac{289}{4}$, or $fx = x - -\frac{13}{2}^2 - \frac{289}{4}$. Therefore, fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that -13/2 and -6.5 are examples of ways to enter a correct answer.

Alternate approach: The graph of $y = fx$ in the *xy*-plane is a parabola. The value of x for the vertex of a parabola is the *x*-value of the midpoint between the two *x*-intercepts of the parabola. Since it's given that $fx = x - 2x + 15$, it follows that the two *x*-intercepts of the graph of $y = fx$ in the *xy*-plane occur when $x = 2$ and $x = -15$, or at the points $2, 0$ and $-15, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2, is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $2, 0$ and $-15, 0$ is $\frac{2 - 15}{2}, \frac{0 + 0}{2}$, or $-\frac{13}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that -13/2 and -6.5 are examples of ways to enter a correct answer.

▼ 3.23 — 3.44, 86/25

The correct answer is 3.44. It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x, this relationship can be represented by an exponential function of the form $fx = ab^x$, where a represents the initial value of the function and b represents the rate of decay,
expressed as a decimal. Because $f0 = 86$, the value of a must be 86. Because the value of fx decreases by 80% for each 1-unit increase in x, the value of b must be $(1 - 0.80)$, or 0.2. Therefore, the function f can be defined by $fx = 860.2^x$. Substituting 2 for x in this function

yields $f2 = 860 . 2^2$, which is equivalent to $f2 = 860 . 04$, or $f2 = 3 . 44$. Either 3.44 or $86 / 25$ may be entered as the correct answer.

Alternate approach: It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Therefore, when $x = 1$, the value of fx is $(100 - 80)\%$, or 20%, of 86, which can be expressed as $0 . 2086$. Since $0 . 2086 = 17 . 2$, the value of $f1$ is 17.2. Similarly, when $x = 2$, the value of fx is 20% of 17.2, which can be expressed as $0 . 2017 . 2$. Since $0 . 2017 . 2 = 3 . 44$, the value of $f2$ is 3.44. Either 3.44 or $86 / 25$ may be entered as the correct answer.

▼ 3.24 — C

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $fx = ax - h^2 + k$, where fx is the height of the object x seconds after it was launched, and a, h, and k are constants such that the function reaches its maximum value, k, when $x = h$. Since the model indicates the object reaches its maximum height of $1{,}034$ feet above the ground 8 seconds after being launched, fx reaches its maximum value, $1{,}034$, when $x = 8$. Therefore, $k = 1{,}034$ and $h = 8$. Substituting 8 for h and $1{,}034$ for k in the function $fx = ax - h^2 + k$ yields $fx = ax - 8^2 + 1{,}034$. Since the model indicates the object has an initial height of 10 feet above the ground, the value of fx is 10 when $x = 0$. Substituting 0 for x and 10 for fx in the equation $fx = ax - 8^2 + 1{,}034$ yields $10 = a0 - 8^2 + 1{,}034$, or $10 = 64a + 1{,}034$. Subtracting $1{,}034$ from both sides of this equation yields $64a = -1{,}024$. Dividing both sides of this equation by 64 yields $a = -16$. Therefore, the model can be represented by the equation $fx = -16x - 8^2 + 1{,}034$. Substituting 10 for x in this equation yields $f10 = -1610 - 8^2 + 1{,}034$, or $f10 = 970$. Therefore, based on the model, 10 seconds after being launched, the height of the object above the ground is 970 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.25 — D

Choice D is correct. It's given that the graph of $y = h(x)$ intersects the x-axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x-axis when $y = 0$ or when

$h(x)=0$, it follows that $h(0)=0$ and $h(t)=0$. If $h(t)=0$, then $0=2(t-4)^2-32$. Adding 32 to both sides of this equation yields $32=2(t-4)^2$. Dividing both sides of this equation by 2 yields $16=(t-4)^2$. Taking the square root of both sides of this equation yields $4=t-4$. Adding 4 to both sides of this equation yields $8=t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

▼ 3.26 — D

Choice D is correct. The equation of a parabola in the *xy*-plane can be written in the form $y = ax - h^2 + k$, where a is a constant and h, k is the vertex of the parabola. If a is positive, the parabola will open upward, and if a is negative, the parabola will open downward. It's given that the parabola has vertex $9, -14$. Substituting 9 for h and -14 for k in the equation $y = ax - h^2 + k$ gives $y = ax - 9^2 - 14$, which can be rewritten as $y = ax - 9x - 9 - 14$, or $y = ax^2 - 18x + 81 - 14$. Distributing the factor of a on the right-hand side of this equation yields $y = ax^2 - 18ax + 81a - 14$. Therefore, the equation of the parabola, $y = ax^2 - 18ax + 81a - 14$, can be written in the form $y = ax^2 + bx + c$, where $a = a$, $b = -18a$, and $c = 81a - 14$. Substituting $-18a$ for b and $81a - 14$ for c in the expression $a + b + c$ yields $a + -18a + 81a - 14$, or $64a - 14$. Since the vertex of the parabola, $9, -14$, is below the *x*-axis, and it's given that the parabola intersects the *x*-axis at two points, the parabola must open upward. Therefore, the constant a must have a positive value. Setting the expression $64a - 14$ equal to the value in choice D yields $64a - 14 = -12$. Adding 14 to both sides of this equation yields $64a = 2$. Dividing both sides of this equation by 64 yields $a = \frac{2}{64}$, which is a positive value. Therefore, if the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants, the value of $a + b + c$ could be -12.

Choice A is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants and $a + b + c = -23$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the *x*-axis at zero points, not two points.

Choice B is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants and $a + b + c = -19$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the *x*-axis at zero points, not two points.

Choice C is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants and $a + b + c = -14$, then the value of a will be 0, which is inconsistent with the equation of a parabola.

▼ 3.27 — A

Choice A is correct. It's given that the graph of $y = fx$ in the xy-plane passes through the points $7, 0$ and $-3, 0$. It follows that when the value of x is either 7 or -3, the value of fx is 0. It's also given that the function f is defined by $fx = ax^2 + bx + c$, where a, b, and c are constants. It follows that the function f is a quadratic function and, therefore, may be written in factored form as $fx = ax - ux - v$, where the value of fx is 0 when x is either u or v. Since the value of fx is 0 when the value of x is either 7 or -3, and the value of fx is 0 when the value of x is either u or v, it follows that u and v are equal to 7 and -3. Substituting 7 for u and -3 for v in the equation $fx = ax - ux - v$ yields $fx = ax - 7x - -3$, or $fx = ax - 7x + 3$. Distributing the right-hand side of this equation yields $fx = ax^2 - 7x + 3x - 21$, or $fx = ax^2 - 4ax - 21a$. Since it's given that $fx = ax^2 + bx + c$, it follows that $b = -4a$. Adding a to each side of this equation yields $a + b = -3a$. Since $a + b = -3a$, if a is an integer, the value of $a + b$ must be a multiple of 3. If a is an integer greater than 1, it follows that $a \geq 2$. Therefore, $-3a \leq -32$. It follows that the value of $a + b$ is less than or equal to -32, or -6. Of the given choices, only -6 is a multiple of 3 that's less than or equal to -6.

Choice B is incorrect. This is the value of $a + b$ if a is equal to, not greater than, 1.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.28 — D

Choice D is correct. According to the table, when x is -1 or 2, $p(x) = 0$. Therefore, two x-intercepts of the graph of p are $(-1,0)$ and $(2,0)$. Since $(-1,0)$ and $(2,0)$ are x-intercepts, it follows that $(x+1)$ and $(x-2)$ are factors of the polynomial equation. This is because when $x = -1$, the value of $x+1$ is 0. Similarly, when $x = 2$, the value of $x-2$ is 0. Therefore, the product $(x+1)(x-2)$ is a factor of the polynomial function p.

Choice A is incorrect. The factor $x-3$ corresponds to an x-intercept of $(3,0)$, which isn't present in the table. Choice B is incorrect. The factor $x+3$ corresponds to an x-intercept of

$(-3,0)$, which isn't present in the table. Choice C is incorrect. The factors $x-1$ and $x+2$ correspond to x-intercepts $(1,0)$ and $(-2,0)$, respectively, which aren't present in the table.

▼ 3.29 — 609

The correct answer is 609. It's given that the function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g15a = \frac{15a}{a} - 14$. This function can be rewritten as $g15a = \frac{15a}{a} - 14$, or $g15a = 15\frac{a}{a} - 14$. Since $a < 0$, it follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g15a = 15\frac{a}{a} - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g7a = \frac{7a}{a} - 14$. This function can be rewritten as $g7a = \frac{7a}{a} - 14$, or $g7a = 7\frac{a}{a} - 14$. Since $a < 0$, it again follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g7a = 7\frac{a}{a} - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or 609.

▼ 3.30 — D

Choice D is correct. The number of bacteria per milliliter is doubling each day. For example, from day 1 to day 2, the number of bacteria increased from 2.5×10^5 to 5.0×10^5. At the end of day 3 there are 10^6 bacteria per milliliter. At the end of day 4, there will be $10^6 \times 2$ bacteria per milliliter. At the end of day 5, there will be $(10^6 \times 2) \times 2$, or $10^6 \times (2^2)$ bacteria per milliliter, and so on. At the end of day d, the number of bacteria will be $10^6 \times (2^{d-3})$. If the number of bacteria per milliliter will reach 5.12×10^8 at the end of day d, then the equation $10^6 \times (2^{d-3}) = 5.12 \times 10^8$ must hold. Since 5.12×10^8 can be rewritten as 512×10^6, the equation is equivalent to $2^{d-3} = 512$. Rewriting 512 as 2^9 gives $d - 3 = 9$, so $d = 12$. The number of bacteria per milliliter would reach 5.12×10^8 at the end of day 12.

Choices A, B, and C are incorrect. Given the growth rate of the bacteria, the number of bacteria will not reach 5.12×10^8 per milliliter by the end of any of these days.

▼ 3.31 — D

Choice D is correct. A function of the form $fx = ab^x + c$, where $a < 0$ and $b > 1$, is a decreasing function. Both of the given functions are of this form; therefore, both are decreasing functions. If a function f is decreasing as the value of x increases, the

corresponding value of fx decreases; therefore, the function doesn't have a minimum value. Thus, neither of the given functions has a minimum value.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.32 — B

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $fx = ax - h^2 + k$, where fx is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. Therefore, $k = 51.1$ and $h = 3$. Substituting 51.1 for k and 3 for h in the equation $fx = ax - h^2 + k$ yields $fx = ax - 3^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground. Therefore, when $x = 0$, $fx = 7$. Substituting 0 for x and 7 for fx in the equation $fx = ax - 3^2 + 51.1$ yields $7 = a0 - 3^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by 9 yields $-4.9 = a$. Substituting -4.9 for a in the equation $fx = ax - 3^2 + 51.1$ yields $fx = -4.9x - 3^2 + 51.1$. Therefore, the equation $fx = -4.9x - 3^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of 7 meters is the value of x when $fx = 7$. Substituting 7 for fx in $fx = -4.9x - 3^2 + 51.1$ yields $7 = -4.9x - 3^2 + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = -4.9x - 3^2$. Dividing both sides of this equation by -4.9 yields $9 = x - 3^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding 3 to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding 3 to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since 0 seconds after the launch represents the time at which the projectile was launched, 6 must be the number of seconds the model estimates that the projectile will return to a height of 7 meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also

given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took 3 seconds to travel from 7 meters above the ground to its maximum height of 51.1 meters above the ground, the model also estimates the projectile will take 3 more seconds to travel from its maximum height of 51.1 meters above the ground back to 7 meters above the ground. Thus, the model estimates that the projectile will return to a height of 7 meters 3 seconds after it reaches its maximum height, which is 6 seconds after the launch.

Choice A is incorrect. The model estimates that 3 seconds after the launch, the projectile reached a height of 51.1 meters, not 7 meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.33 — 14

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

▼ 3.34 — 1728

The correct answer is 1,728. The given function can be rewritten in the form $fx = ax - h^2 + k$, where a is a positive constant and the minimum value, k, of the function occurs when the value of x is h. By completing the square, $fx = x^2 - 48x + 2{,}304$ can be written as $fx = x^2 - 48x + \frac{48}{2}^2 + 2{,}304 - \frac{48}{2}^2$, or $fx = x - 24^2 + 1{,}728$. This equation is in the form $fx = ax - h^2 + k$, where $a = 1$, $h = 24$, and $k = 1{,}728$. Therefore, the minimum value of the given function is 1,728.

▼ 3.35 — 44

The correct answer is 44. For a function defined by an equation of the form $fx = a1 + \frac{r}{100}^{\frac{x}{k}}$, where a, r, and k are positive constants, the value of fx increases by r% for every increase of x by k. The given equation can be rewritten as $fx = 271.20^{\frac{2x}{6}}$, or $fx = 271.2^{2\,\frac{x}{6}}$, which is equivalent to $fx = 271.44^{\frac{x}{6}}$, or $fx = 271 + \frac{44}{100}^{\frac{x}{6}}$. This equation is of the form $fx = a1 + \frac{r}{100}^{\frac{x}{k}}$, where $a = 27$, $r = 44$, and $k = 6$. It follows that the value of fx increases by 44% for every increase of x by 6. Therefore, the value of p is 44.

Alternate approach: It's given that $fx = 271.20^{\frac{x}{3}}$. An increase in the value of x by 6 can be represented by the expression $x + 6$. The value of $fx + 6$ can be found by substituting $x + 6$ for x in the equation $fx = 271.20^{\frac{x}{3}}$, which yields $fx + 6 = 271.20^{\frac{x+6}{3}}$. By the product property of exponents, this equation is equivalent to $fx + 6 = 271.20^{\frac{x}{3}} \cdot 1.20^{\frac{6}{3}}$, or $fx + 6 = 271.20^{\frac{x}{3}} \cdot 1.20^2$, which can be rewritten as $fx + 6 = 271.20^{\frac{x}{3}} \cdot 1.44$. Since $fx = 271.20^{\frac{x}{3}}$, it follows that $f(x + 6) = f(x) \cdot (1.44)$. Therefore, for any value of x, the value of $f(x + 6)$ is 1.44 times, or 44% greater than, the value of fx. It's given that the value of fx increases by p% for every increase in the value of x by 6. It follows that the value of p is 44.

▼ 3.36 — B

Choice B is correct. A cube has 6 faces of equal area, so if the total surface area of a cube is $6\left(\frac{a}{4}\right)^2$, then the area of one face is $\left(\frac{a}{4}\right)^2$. Likewise, the area of one face of a cube is the square of one of its edges; therefore, if the area of one face is $\left(\frac{a}{4}\right)^2$, then the length of one edge of the cube is $\frac{a}{4}$. Since the perimeter of one face of a cube is four times the length of one edge, the perimeter is $4\left(\frac{a}{4}\right) = a$.

Choice A is incorrect because if the perimeter of one face of the cube is $\frac{a}{4}$, then the total

surface area of the cube is $6\left(\frac{\frac{a}{4}}{4}\right)^2 = 6\left(\frac{a}{16}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice C is incorrect because if the perimeter of one face of the cube is 4a, then the total surface area of the cube is $6\left(\frac{4a}{4}\right)^2 = 6a^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice D is incorrect because if the perimeter of one face of the cube is 6a, then the total surface area of the cube is $6\left(\frac{6a}{4}\right)^2 = 6\left(\frac{3a}{2}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

▼ 3.37 — -12

The correct answer is -12. It's given that function *f* is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the *xy*-plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the *x*-coordinate of the vertex of the graph is the average of *a* and *b*. Therefore, the *x*-coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20 + -4}{2}$, or -12. It's given that the graph of $y = f(x)$ in the *xy*-plane has a vertex at $(r, -64)$. It follows that the value of *r* is -12.

▼ 3.38 — D

Choice D is correct. A *y*-intercept of a graph in the *xy*-plane is a point where the graph intersects the *y*-axis, or a point where $x = 0$. Substituting 0 for *x* in the equation defining function *f* yields $f0 = a2.2^0 + 2.2^b$, or $f0 = a1 + 2.2^b$. So, the *y*-coordinate of the *y*-intercept of the graph is $a1 + 2.2^b$, or equivalently, $a + a2.2^b$. It's given that function *g* is equivalent to function *f*, where $0 < a < b$. It follows that $k = 2.2^b$. Since $a2.2^b$ can't be equal to 0, the coefficient *a* can't be equal to $a + a2.2^b$. Since $0 < a$, the constant *k*, which is equal to 2.2^b, can't be equal to $a + a2.2^b$. Therefore, function *g* doesn't display the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane as a constant or coefficient. It's also given that function *h* is equivalent to function *f*, where $0 < a < b$. The equation defining *f* can be rewritten as $fx = a2.2^x + a2.2^b$. It follows that $m = a2.2^b$. Since $a2.2^b$ can't be equal to 0, the coefficient *a* can't be equal to $a + a2.2^b$. Since $0 < a$, the constant *m*, which is equal to $a2.2^b$, can't be equal to $a + a2.2^b$. Therefore, function *h* doesn't display the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane as a constant or coefficient. Thus, neither function *g* nor function *h* displays the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane as a constant or coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.39 — 336

The correct answer is 336. By the zero product property, if $x + 14t - x = 0$, then $x + 14 = 0$, which gives $x = -14$, or $t - x = 0$, which gives $x = t$. Therefore, $gx = 0$ when $x = -14$ and when $x = t$. Since the graph of $y = gx$ passes through the point $24, 0$, it follows that $g24 = 0$, so $t = 24$. Substituting 24 for t in the equation $gx = x + 14t - x$ yields $gx = x + 1424 - x$. The value of $g0$ can be calculated by substituting 0 for x in this equation, which yields $g0 = 0 + 1424 - 0$, or $g0 = 336$.

▼ 3.40 — A

Choice A is correct. It's given that $gx = fx + 5$. Since $fx = 4x^2 + 64x + 262$, it follows that $fx + 5 = 4x + 5^2 + 64x + 5 + 262$. Expanding the quantity $x + 5^2$ in this equation yields $fx + 5 = 4x^2 + 10x + 25 + 64x + 5 + 262$. Distributing the 4 and the 64 yields $fx + 5 = 4x^2 + 40x + 100 + 64x + 320 + 262$. Combining like terms yields $fx + 5 = 4x^2 + 104x + 682$. Therefore, $gx = 4x^2 + 104x + 682$. For a quadratic function defined by an equation of the form $gx = ax - h^2 + k$, where a, h, and k are constants and a is positive, gx reaches its minimum, k, when the value of x is h. The equation $gx = 4x^2 + 104x + 682$ can be rewritten in this form by completing the square. This equation is equivalent to $gx = 4x^2 + 26 + 682$, or $gx = 4x^2 + 26x + 169 - 169 + 682$. This equation can be rewritten as $gx = 4x + 13^2 - 169 + 682$, or $gx = 4x + 13^2 - 4169 + 682$, which is equivalent to $gx = 4x + 13^2 + 6$. This equation is in the form $gx = ax - h^2 + k$, where $a = 4$, $h = -13$, and $k = 6$. Therefore, gx reaches its minimum when the value of x is -13.

Choice B is incorrect. This is the value of x for which fx, rather than gx, reaches its minimum.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $fx - 5$, rather than $fx + 5$, reaches its minimum.

▼ 3.41 — D

Choice D is correct. It's given that an exponential model estimates that the number of comments on an article increased by a fixed percentage at the end of each hour. Therefore, the model can be represented by an exponential equation of the form $C = Ka^t$, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and K and a are constants. It's also given that when the article was first featured on the home page of the news website, there were 40 comments on the article. This means that when $t = 0$, $C = 40$. Substituting 0 for t and 40 for C in the equation $C = Ka^t$ yields $40 = Ka^0$, or $40 = K$. It's also given that the number of comments on the article at the end of an hour had increased by 190% of the number of comments on the article at the end of the previous hour. Multiplying the percent increase by the number of comments on the article at the end of the previous hour yields the number of estimated additional comments the article has on its home page: $40\frac{190}{100}$, or 76 comments. Thus, the estimated number of comments for the following hour is the sum of the comments from the end of the previous hour and the number of additional comments, which is $40 + 76$, or 116. This means that when $t = 1$, $C = 116$. Substituting 1 for t, 116 for C, and 40 for K in the equation $C = Ka^t$ yields $116 = 40a^1$, or $116 = 40a$. Dividing both sides of this equation by 40 yields $2.9 = a$. Substituting 40 for K and 2.9 for a in the equation $C = Ka^t$ yields $C = 402.9^t$. Thus, the equation that best represents this model is $C = 40(2.9)^t$.

Choice A is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 19% of the number of comments at the end of the previous hour, rather than 190%.

Choice B is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 90% of the number of comments at the end of the previous hour, rather than 190%.

Choice C is incorrect. This model represents a situation where the number of comments at the end of each hour was 19 times the number of comments at the end of the previous hour, rather than increasing by 190% of the number of comments at the end of the previous hour.

▼ 3.42 — -2112

The correct answer is $-2{,}112$. It's given that f is a quadratic function. It follows that f can be defined by an equation of the form $fx = ax - h^2 + k$, where a, h, and k are constants. It's also given that the table shows three values of x and their corresponding values of y, where $y = fx + 4$. Substituting $ax - h^2 + k$ for fx in this equation yields $y = ax - h^2 + k + 4$. This equation represents a quadratic relationship between x and y, where $k + 4$ is either the

maximum or the minimum value of y, which occurs when $x = h$. For quadratic relationships between x and y, the maximum or minimum value of y occurs at the value of x halfway between any two values of x that have the same corresponding value of y. The table shows that x-values of 21 and 25 correspond to the same y-value, -8. Since 23 is halfway between 21 and 25, the maximum or minimum value of y occurs at an x-value of 23. The table shows that when $x = 23$, $y = 8$. It follows that $h = 23$ and $k + 4 = 8$. Subtracting 4 from both sides of the equation $k + 4 = 8$ yields $k = 4$. Substituting 23 for h and 4 for k in the equation $y = ax - h^2 + k + 4$ yields $y = ax - 23^2 + 4 + 4$, or $y = ax - 23^2 + 8$. The value of a can be found by substituting any x-value and its corresponding y-value for x and y, respectively, in this equation. Substituting 25 for x and -8 for y in this equation yields $-8 = a25 - 23^2 + 8$, or $-8 = a2^2 + 8$. Subtracting 8 from both sides of this equation yields $-16 = a2^2$, or $-16 = 4a$. Dividing both sides of this equation by 4 yields $-4 = a$. Substituting -4 for a, 23 for h, and 4 for k in the equation $fx = ax - h^2 + k$ yields $fx = -4x - 23^2 + 4$. The y-intercept of the graph of $y = fx$ in the xy-plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $fx = -4x - 23^2 + 4$ yields $f0 = -40 - 23^2 + 4$, or $f0 = -4-23^2 + 4$. This is equivalent to $f0 = -2{,}112$, so the y-intercept of the graph of $y = fx$ in the xy-plane is $0, -2{,}112$. Thus, the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane is $-2{,}112$.

▼ 3.43 — B

Choice B is correct. Functions f and g are both exponential functions with a base of 0.40. Since 0.40 is less than 1, functions f and g are both decreasing exponential functions. This means that fx and gx decrease as x increases. Since fx and gx decrease as x increases, the maximum value of each function occurs at the least value of x for which the function is defined. It's given that functions f and g are defined for $x \geq 0$. Therefore, the maximum value of each function occurs at $x = 0$. Substituting 0 for x in the equation defining f yields $f0 = 330.4^{0+3}$, which is equivalent to $f0 = 330.4^3$, or $f0 = 2.112$. Therefore, the maximum value of f is 2.112. Since the equation $fx = 330.4^{x+3}$ doesn't display the value 2.112, the equation defining f doesn't display the maximum value of f. Substituting 0 for x in the equation defining g yields $g0 = 330.160.4^{0-2}$, which can be rewritten as $g0 = 330.16\frac{1}{0.4^2}$, or $g0 = 330.16\frac{1}{0.16}$, which is equivalent to $g0 = 33$. Therefore, the maximum value of g is 33. Since the equation $gx = 330.160.4^{x-2}$ displays the value 33, the equation defining g displays the maximum value of g. Thus, only equation II displays, as a constant or coefficient, the maximum value of the function it defines.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.44 — A

Choice A is correct. It's given that the function P models the population, in thousands, of a certain city t years after 2003. The value of the base of the given exponential function, 1.04, corresponds to an increase of 4% for every increase of 1 in the exponent, $\frac{6}{4}t$. If the exponent is equal to 0, then $\frac{6}{4}t = 0$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = 0$. If the exponent is equal to 1, then $\frac{6}{4}t = 1$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = \frac{4}{6}$, or $t = \frac{2}{3}$. Therefore, the population is predicted to increase by 4% every $\frac{2}{3}$ of a year. It's given that the population is predicted to increase by 4% every n months. Since there are 12 months in a year, $\frac{2}{3}$ of a year is equivalent to $\frac{2}{3}12$, or 8, months. Therefore, the value of n is 8.

Choice B is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice C is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{4}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice D is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{1}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

▼ 3.45 — D

Choice D is correct. An equation representing the height above ground h, in meters, of a softball t seconds after it is launched by a machine from ground level can be written in the form $h = -a(t - b)^2 + c$, where a, b, and c are positive constants. In this equation, b represents the time, in seconds, at which the softball reaches its maximum height of c meters above the ground. It's given that this softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds; therefore, $b = 1.8$ and $c = 51.84$. Substituting 1.8 for b and 51.84 for c in the equation $h = -a(t - b)^2 + c$ yields $h = -a(t - 1.8)^2 + 51.84$. It's also given that this softball hits the ground at 3.6 seconds; therefore, $h = 0$ when $t = 3.6$. Substituting 0 for h and 3.6 for t in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $0 = -a(3.6 - 1.8)^2 + 51.84$, which is equivalent to $0 = -a(1.8)^2 + 51.84$, or $0 = -3.24a + 51.84$. Adding $3.24a$ to both sides of this equation yields $3.24a = 51.84$. Dividing both sides of this equation by 3.24 yields $a = 16$. Substituting 16 for a in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $h = -16(t - 1.8)^2 + 51.84$. Therefore, $h = -16(t - 1.8)^2 + 51.84$ represents the height above ground h, in meters, of this softball t seconds after it is launched.

Choice A is incorrect. This equation represents a situation where the maximum height is 3.6 meters above the ground at 0 seconds, not 51.84 meters above the ground at 1.8 seconds.

Choice B is incorrect. This equation represents a situation where the maximum height is 51.84 meters above the ground at 0 seconds, not 1.8 seconds.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.46 — C

Choice C is correct. It's given that the function f gives the product of a number, x, and a number that is 91 more than x. A number that is 91 more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.47 — A

Choice A is correct. The y-intercept of the graph of $y = fx$ in the xy-plane occurs at the point on the graph where $x = 0$. In other words, when $x = 0$, the corresponding value of fx is the y-coordinate of the y-intercept. Substituting 0 for x in the given equation yields $f0 = -82^0 + 22$, which is equivalent to $f0 = -81 + 22$, or $f0 = 14$. Thus, when $x = 0$, the corresponding value of fx is 14. Therefore, the y-intercept of the graph of $y = fx$ in the xy-plane is $0, 14$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be the y-intercept for $fx = -82^x$, not $fx = -82^x + 22$.

▼ 3.48 — C

Choice C is correct. By definition, the y-value when a function intersects the x-axis is 0. It's given that the graph of the function intersects the x-axis at 0 and a, that x is the unit price, in dollars, of a product, and that $f(x)$, where $y = f(x)$, is the revenue, in dollars, that a

company receives from the sales of the product. Since the value of a occurs when $y = 0$, a is the unit price, in dollars, of the product that will result in a revenue of $0.

Choice A is incorrect. The revenue, in dollars, when the unit price of the product is $0 is represented by $f(x)$, when $x = 0$. Choice B is incorrect. The unit price, in dollars, of the product that will result in maximum revenue is represented by the x-coordinate of the maximum of f. Choice D is incorrect. The maximum revenue, in dollars, that the company can make is represented by the y-coordinate of the maximum of f.

▼ 3.49 — C

Choice C is correct. It's given that equation $y = ab^{\frac{x}{100}}$ represents the model, where y is the estimated volume, in cubic centimeters, of a certain tree in a forest and x is the tree's height, in centimeters. For an exponential equation of this form to represent a model where the value of y increases as the value of x increases, it must follow that the values of a and b are positive. It's also given that the tree's estimated volume increases by 13.9% for every increase of 100 centimeters in the tree's height. Therefore, for every 100-centimeter increase in the tree's height, the estimated volume of the tree is 1.139 times larger. In other words, $ab^{\frac{x+100}{100}} = 1.139ab^{\frac{x}{100}}$, which is equivalent to $ab^{\frac{x}{100}}b^{\frac{100}{100}} = 1.139ab^{\frac{x}{100}}$. Since a and b are positive, dividing both sides of this equation by $ab^{\frac{x}{100}}$ yields $b^{\frac{100}{100}} = 1.139$, or $b = 1.139$. Therefore, the value of b is 1.139.

Choice A is incorrect. This is the given percent increase in decimal form, not the value of b.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the given percent increase, not the value of b.

▼ 3.50 — ?

The correct answer is 8. The values of $f(x)$ for the exponential function f shown in the table increase by a factor of a^4 for each increase of 1 in x. This relationship can be represented by the equation $f(x) = a^{4x+b}$, where b is a constant. It's given that when $x = 2$, $f(x) = a^5$. Substituting 2 for x and a^5 for $f(x)$ into $f(x) = a^{4x+b}$ yields $a^5 = a^{4(2)+b}$. Since $4(2)+b = 5$, it follows that $b = -3$. Thus, an equation that defines the function f is $f(x) = a^{4x-3}$. It

follows that the value of k such that $f(k) = a^{29}$ can be found by solving the equation $4k - 3 = 29$, which yields $k = 8$.

▼ 3.51 — B

Choice B is correct. The graph of a quadratic function in the xy-plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x-intercepts of the graph have the same x-coordinate. Since $f(-3) = f(-1) = 0$, the x-coordinate of the vertex is $\frac{(-3)+(-1)}{2} = -2$. Of the shown intervals, only the interval in choice B contains –2. Choices A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x-intercepts.

▼ 3.52 — A

Choice A is correct. The equation $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1.84^{\frac{1}{4}x}$, which is equivalent to $fx = 1.84^{\frac{1}{4}^{x}}$, or approximately $fx = 1.16467^{x}$. Since it's given that $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1 + \frac{p}{100}^{x}$, where p is a constant, it follows that $1 + \frac{p}{100}$ is approximately equal to 1.16467. Therefore, $\frac{p}{100}$ is approximately equal to 0.16467. It follows that the value of p is approximately equal to 16.467. Of the given choices, 16 is closest to the value of p.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.53 — A

Choice A is correct. It's given that the table shows values of x and their corresponding values of gx, where $gx = \frac{fx}{x+3}$. It's also given that f is a linear function. It follows that an equation that defines f can be written in the form $fx = mx + b$, where m represents the slope and b represents the y-coordinate of the y-intercept $0, b$ of the graph of $y = fx$ in the xy-plane. The slope of the graph of $y = fx$ can be found using two points, x_1, y_1 and x_2, y_2, that are on the graph of $y = fx$, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since the table shows values of x and their corresponding values of gx, substituting values of x and gx in the equation $gx = \frac{fx}{x+3}$ can be used to define function f. Using the first pair of values from the table, $x = -27$ and $gx = 3$, yields $3 = \frac{f-27}{-27+3}$, or $3 = \frac{f-27}{-24}$. Multiplying each side of this equation by -24 yields $-72 = f-27$, so

the point $-27, -72$ is on the graph of $y = fx$. Using the second pair of values from the table, $x = -9$ and $gx = 0$, yields $0 = \frac{f-9}{-9+3}$, or $0 = \frac{f-9}{-6}$. Multiplying each side of this equation by -6 yields $0 = f-9$, so the point $-9, 0$ is on the graph of $y = fx$. Substituting $-27, -72$ and $-9, 0$ for x_1, y_1 and x_2, y_2, respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{0 - -72}{-9 - -27}$, or $m = 4$. Substituting 4 for m in the equation $fx = mx + b$ yields $fx = 4x + b$. Since $0 = f-9$, substituting -9 for x and 0 for fx in the equation $fx = 4x + b$ yields $0 = 4-9 + b$, or $0 = -36 + b$. Adding 36 to both sides of this equation yields $36 = b$. It follows that 36 is the y-coordinate of the y-intercept $0, b$ of the graph of $y = fx$. Therefore, the y-intercept of the graph of $y = fx$ is $0, 36$.

Choice B is incorrect. 12 is the y-coordinate of the y-intercept of the graph of $y = gx$.

Choice C is incorrect. 4 is the slope of the graph of $y = fx$.

Choice D is incorrect. -9 is the x-coordinate of the x-intercept of the graph of $y = fx$.

▼ 3.54 — 48

The correct answer is 48. It's given that in the xy-plane, when the graph of the function f, where $y = fx$, is shifted up 6 units, the resulting graph is defined by the function g. Therefore, function g can be defined by the equation $gx = fx + 6$. It's given that $fx = x - 1x + 3x - 2$. Substituting $x - 1x + 3x - 2$ for fx in the equation $gx = fx + 6$ yields $gx = x - 1x + 3x - 2 + 6$. For the point $4, b$, the value of x is 4. Substituting 4 for x in the equation $gx = x - 1x + 3x - 2 + 6$ yields $g4 = 4 - 14 + 34 - 2 + 6$, or $g4 = 48$. It follows that the graph of $y = gx$ crosses through the point $4, 48$. Therefore, the value of b is 48.

▼ 3.55 — C

Choice C is correct. It's given that function f is a quadratic function, and in the xy-plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$. In the xy-plane, the graph of a quadratic function with a vertex at (h, k) has an equation of the form $y = a(x - h)^2 + k$, where a is a constant. Therefore, function f can be written as $fx = a(x - 1)^2 + 9$. It's also given that the graph of $y = f(x)$ passes through the point $(2, 33)$. Substituting 2 for x and 33 for $f(x)$ in the equation $fx = a(x - 1)^2 + 9$ yields $33 = a(2 - 1)^2 + 9$, or $33 = a + 9$. Subtracting 9 from both sides of this equation yields $24 = a$. Substituting 24 for a in the equation $fx = a(x - 1)^2 + 9$ yields $f(x) = 24(x - 1)^2 + 9$. The value of $f(-2)$ can be found by substituting -2 for x in function f. Therefore, $f-2 = 24(-2 - 1)^2 + 9$, which yields $f(-2) = 24(9) + 9$, or $f(-2) = 225$. The value of $f(0)$ can be found by substituting 0 for x in function f. Therefore, $f0 = 24(0 - 1)^2 + 9$, which yields $f(0) = 24 + 9$, or $f(0) = 33$. It follows that $f(-2) - f(0) = 225 - 33$, or $f(-2) - f(0) = 192$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f(-2)$, not the value of $f(-2) - f(0)$.

▼ 3.56 — D

Choice D is correct. Since the value of y increases by a constant factor, 4, for each increase of 1 in the value of x, the relationship between x and y is exponential. An exponential relationship between x and y can be represented by an equation of the form $y = ab^x$, where a is the value of x when $y = 0$ and y increases by a factor of b for each increase of 1 in the value of x. Since $y = 200$ when $x = 0$, $a = 200$. Since y increases by a factor of 4 for each increase of 1 in the value of x, $b = 4$. Substituting 200 for a and 4 for b in the equation $y = ab^x$ yields $y = 200(4)^x$. Thus, the equation $y = 200(4)^x$ represents the relationship between x and y.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This equation represents a relationship where for each increase of 1 in the value of x, the value of y increases by a factor of 200, not 4, and when $x = 0$, y is equal to 4, not 200.

Choice C is incorrect and may result from conceptual errors.

▼ 3.57 — 5

The correct answer is 5. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 15$ yields $y = -a^x + b - 15$. It's given that the y-intercept of the graph of $y = fx - 15$ is $0, -\frac{99}{7}$. Substituting 0 for x and $-\frac{99}{7}$ for y in the equation $y = -a^x + b - 15$ yields $-\frac{99}{7} = -a^0 + b - 15$, which is equivalent to $-\frac{99}{7} = -1 + b - 15$, or $-\frac{99}{7} = b - 16$. Adding 16 to both sides of this equation yields $\frac{13}{7} = b$. It's given that the product of a and b is $\frac{65}{7}$, or $ab = \frac{65}{7}$. Substituting $\frac{13}{7}$ for b in this equation yields $a\frac{13}{7} = \frac{65}{7}$. Dividing both sides of this equation by $\frac{13}{7}$ yields $a = 5$.

▼ 3.58 — A

Choice A is correct. The given quadratic function f is in vertex form, $f(x) = (x-h)^2 + k$, where (h,k) is the vertex of the graph of $y = f(x)$ in the xy-plane. Therefore, the vertex of the graph of $y = f(x)$ is $(2,-4)$. In addition, the y-coordinate of the vertex represents either the minimum or maximum value of a quadratic function, depending on whether the graph of the function opens upward or downward. Since the leading coefficient of f (the coefficient of the

term $(x-2)^2$) is 1, which is positive, the graph of $y = f(x)$ opens upward. It follows that at $x = 2$, the minimum value of the function f is -4.

Choice B is incorrect and may result from making a sign error and from using the x-coordinate of the vertex. Choice C is incorrect and may result from using the x-coordinate of the vertex. Choice D is incorrect and may result from making a sign error.

▼ 3.59 — D

Choice D is correct. The *y*-intercept of a graph in the *xy*-plane is the point where $x = 0$. For the given function *f*, the *y*-intercept of the graph of $y = fx$ in the *xy*-plane can be found by substituting 0 for *x* in the equation $y = 9{,}0000.66^x$, which gives $y = 9{,}0000.66^0$. This is equivalent to $y = 9{,}0001$, or $y = 9{,}000$. Therefore, the *y*-intercept of the graph of $y = fx$ is $0, 9{,}000$. It's given that the function *f* models the number of advertisements a company sent to its clients each year. Therefore, *fx* represents the estimated number of advertisements the company sent to its clients each year. It's also given that *x* represents the number of years since 1997. Therefore, $x = 0$ represents 0 years since 1997, or 1997. Thus, the best interpretation of the *y*-intercept of the graph of $y = fx$ is that the estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.60 — D

Choice D is correct. A function defined by an equation in the form $fx = ab^{x+h}$, where *a*, *b*, and *h* are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function *g* is defined by $gx = 181.161.4^{x+2}$, which is equivalent to $gx = 20.881.4^{x+2}$. Substituting 0 for *x* in this equation yields $g0 = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $gx = 181.161.4^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function *h* is defined by $hx = 181.4^{x+4}$. Substituting 0 for *x* in this equation yields $h0 = 181.4^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $hx = 181.4^{x+4}$ doesn't display its minimum value as a constant or coefficient.

Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.61 — C

Choice C is correct. For a function of the form $fx = ar^{\frac{x}{k}}$, where a, r, and k are constants and $r < 1$, the value of fx decreases by $1001 - r\%$ for every increase of x by k. In the given function, $a = 5{,}470$, $r = 0.64$, and $k = 12$. Therefore, for the given function, the value of fx decreases by $1001 - 0.64\%$, or 36%, for every increase of x by 12. Since fx represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every 12 months by 36% of its value the preceding 12 months. Since there are 12 months in a year, the value of the equipment decreases each year by 36% of its value the preceding year. Thus, the value of p is 36.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.62 — 20

The correct answer is 20. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 12$ yields $y = -a^x + b - 12$. It's given that the y-intercept of the graph of $y = fx - 12$ is $0, -\frac{75}{7}$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $a\frac{16}{7} = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

▼ 3.63 — D

Choice D is correct. Stating that the population will increase each year by 3% from the previous year is equivalent to saying that the population each year will be 103% of the population the year before. Since the initial population is 50,000, the population after

t years is given by $50{,}000(1.03)^t$. It follows that the equation $60{,}000 = 50{,}000(1.03)^t$ can be used to estimate the number of years it will take for the population to reach 60,000.

Choice A is incorrect. This equation models how long it will take the population to decrease from 60,000 to 50,000, which is impossible given the growth factor. Choice B is incorrect and may result from misinterpreting a 3% growth as growth by a factor of 3. Additionally, this equation attempts to model how long it will take the population to decrease from 60,000 to 50,000. Choice C is incorrect and may result from misunderstanding how to model percent growth by multiplying the initial amount by a factor greater than 1.

▼ 3.64 — A

Choice A is correct. It's given that in the xy-plane, the graph of $y = g(x)$ passes through the point $(0, 22)$. It follows that $g(0) = 22$. Substituting 0 for x and 22 for $g(x)$ in the given equation yields $22 = \frac{0^2 - 0 - a}{0^3 - 0 - b}$, which is equivalent to $22 = \frac{-a}{-b}$, or $22 = \frac{a}{b}$. Multiplying each side of this equation by b yields $22b = a$. Substituting $22b$ for a in the equation defining the function g yields $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$. It's given that $g(-22) = 0$. Substituting -22 for x and 0 for $g(x)$ in the equation $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$ yields $0 = \frac{(-22)^2 - (-22) - 22b}{(-22)^3 - (-22) - b}$. Since a rational expression equals 0 if and only if its numerator is 0, it follows that $(-22)^2 - -22 - 22b = 0$, or $22^2 + 22 - 22b = 0$. Dividing each side of this equation by 22 yields $22 + 1 - b = 0$, or $23 - b = 0$. Adding b to each side of this equation yields $23 = b$, so the value of b is 23.

Choice B is incorrect. This is the value of $g(0)$, not the value of b.

Choice C is incorrect. This is a value of x for which $g(x) = 0$, not the value of b.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.65 — B

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x-coordinate of the vertex of the graph of $y = fx$ is the x-coordinate of the midpoint of its two x-intercepts. The midpoint of two points with x-coordinates x_1 and x_2 has x-coordinate x_m, where $x_m = \frac{x_1 + x_2}{2}$. It's given that the vertex is $-3, 6$ and one of the x-intercepts is $-\frac{17}{4}, 0$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1 + x_2}{2}$ yields $-3 = \frac{-\frac{17}{4} + x_2}{2}$. Multiplying each side of this equation by 2 yields

$-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x-intercept is $-\frac{7}{4}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.66 — D

Choice D is correct. Substituting b for x in the given equation yields $g(b + 28) = \frac{1}{10}(b - a)^2 (b - b)^2 (b - c)$, which is equivalent to $g(b + 28) = \frac{1}{10}(b - a)^2 (0)^2 (b - c)$, or $g(b + 28) = 0$. It follows that when $x = b + 28$, $g(x) = 0$. There are three points where the graph of $y = g(x)$ intersects the x-axis: $(-2, 0)$, $(1, 0)$, and $(3, 0)$. Therefore, $g(-2) = 0$, $g(1) = 0$, and $g(3) = 0$. It follows that either $b + 28 = -2$, $b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields $b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting 28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be either -30, -27, or -25. Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Choice C is incorrect. This would be the value of c, not the value of b, if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

▼ 3.67 — 1.1, 11/10

The correct answer is 1.1. It's given that $h(t) = -16t^2 + b$ estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. It's also given that the function estimates that the ball is 19.36 feet above the floor when it is released at $t = 0$. Substituting 19.36 for $h(t)$ and 0 for t in the given function yields $19.36 = -16(0)^2 + b$, or $19.36 = b$. Substituting 19.36 for b in function h yields $h(t) = -16t^2 + 19.36$. When the ball reaches the floor, its height is 0 feet above the floor. Substituting 0 for $h(t)$ yields $0 = -16t^2 + 19.36$. Adding $16t^2$ to both sides of this equation yields $16t^2 = 19.36$. Dividing both sides of this equation by 16 yields $t^2 = 1.21$. Taking the positive square root of both sides of this equation yields $t = 1.1$. Therefore, the function estimates that the ball will reach

the floor 1.1 seconds after being released. Note that 1.1 and 11/10 are examples of ways to enter a correct answer.

▼ 3.68 — B

Choice B is correct. It's given that $gx = fx - 1$. Since $fx = x + 6x + 5x + 1$, it follows that $fx - 1 = x - 1 + 6x - 1 + 5x - 1 + 1$. Combining like terms yields $fx - 1 = x + 5x + 4x$. Therefore, $gx = xx + 5x + 4$. The x-intercepts of a graph in the xy-plane are the points where $y = 0$. The x-coordinates of the x-intercepts of the graph of $y = gx$ in the xy-plane can be found by solving the equation $0 = xx + 5x + 4$. Applying the zero product property to this equation yields three equations: $x = 0$, $x + 5 = 0$, and $x + 4 = 0$. Solving each of these equations for x yields $x = 0$, $x = -5$, and $x = -4$, respectively. Therefore, the x-intercepts of the graph of $y = gx$ are $0, 0$, $-5, 0$, and $-4, 0$. It follows that the values of a, b, and c are 0, -5, and -4. Thus, the value of $a + b + c$ is $0 + -5 + -4$, which is equal to -9.

Choice A is incorrect. This is the value of $a + b + c$ if $gx = fx + 1$.

Choice C is incorrect. This is the value of $a + b + c - 1$ if $gx = x - 6x - 5x - 1$.

Choice D is incorrect. This is the value of $a + b + c$ if $fx = x - 6x - 5x - 1$.

▼ 3.69 — -4.9, -49/10

The correct answer is -4.9. The value of $f-4$ is the value of fx when $x = -4$. Substituting -4 for x in the equation $fx = 3 + -2x - x^2$ yields $f-4 = 3 + -2-4 - -4^2$, or $f-4 = 3 + -8$, which is equivalent to $f-4 = 3 + 8$, or $f-4 = 11$. Since it's given that $f-4 = c$, it follows that $c = 11$ and the value of gc is the value of $g11$. Substituting 11 for w in the equation $gw = \frac{-w}{w - 1} - w + 5$ yields $g11 = \frac{-11}{11 - 1} - 11 + 5$, or $g11 = -1.1 - 6$, which is equivalent to $g11 = 1.1 - 6$, or $g11 = -4.9$. Note that -4.9 and -49/10 are examples of ways to enter a correct answer.

▼ 3.70 — B

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, 150%, each year, the model can be represented by an exponential equation of the form $n = a1 + \frac{p}{100}^t$, where a is the estimated number of squirrels in the population at the end of 2015, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was 150% more than the number at the end of the previous year, $p = 150$. Substituting 150 for p in the equation $n = a1 + \frac{p}{100}^t$ yields $n = a1 + \frac{150}{100}^t$, which is equivalent to $n = a1 + 1.5^t$, or $n = a2.5^t$. It's given

that the estimated number of squirrels at the end of 2016 was 180. This means that when $t = 1$, $n = 180$. Substituting 1 for t and 180 for n in the equation $n = a2.5^t$ yields $180 = a2.5^1$, or $180 = 2.5a$. Dividing each side of this equation by 2.5 yields $72 = a$. Substituting 72 for a in the equation $n = a2.5^t$ yields $n = 722.5^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

▼ 3.71 — A

Choice A is correct. It's given that the banner's length x, in inches, is 24 inches longer than its width, in inches. It follows that the banner's width, in inches, can be represented by the expression $x - 24$. The area of a rectangle is the product of its length and its width. It's given that the area of the banner is 2,661 square inches, so it follows that $2{,}661 = xx - 24$, or $2{,}661 = x^2 - 24x$. Subtracting 2,661 from each side of this equation yields $0 = x^2 - 24x - 2{,}661$. Therefore, the equation that represents this situation is $0 = x^2 - 24x - 2{,}661$.

Choice B is incorrect and may result from representing the width, in inches, of the banner as $24 - x$, rather than $x - 24$.

Choice C is incorrect and may result from representing the width, in inches, of the banner as $x + 24$, rather than $x - 24$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.72 — D

Choice D is correct. It's given that $f24 < 0$. Substituting 24 for fx in the equation $fx = a\sqrt{x + b}$ yields $f24 = a\sqrt{24 + b}$. Therefore, $a\sqrt{24 + b} < 0$. Since $\sqrt{24 + b}$ can't be negative, it follows that $a < 0$. It's also given that the graph of $y = fx$ passes through the point $-24, 0$. It follows that when $x = -24$, $fx = 0$. Substituting -24 for x and 0 for fx in the equation $fx = a\sqrt{x + b}$ yields

$0 = a\sqrt{-24 + b}$. By the zero product property, either $a = 0$ or $\sqrt{-24 + b} = 0$. Since $a < 0$, it follows that $\sqrt{24 + b} = 0$. Squaring both sides of this equation yields $-24 + b = 0$. Adding 24 to both sides of this equation yields $b = 24$. Since $a < 0$ and b is 24, it follows that $a < b$ must be true.

Choice A is incorrect. The value of $f0$ is $a\sqrt{b}$, which must be negative.

Choice B is incorrect. The value of $f0$ is $a\sqrt{b}$, which could be -24, but doesn't have to be.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 3.73 — 76

The correct answer is 76. It's given that the graph of $y = gx$ is the result of translating the graph of $y = fx$ up 4 units in the xy-plane. It follows that the graph of $y = gx$ is the same as the graph of $y = fx + 4$. Substituting gx for y in the equation $y = fx + 4$ yields $gx = fx + 4$. It's given that $fx = x - 6x - 2x + 6$. Substituting $x - 6x - 2x + 6$ for fx in the equation $gx = fx + 4$ yields $gx = x - 6x - 2x + 6 + 4$. Substituting 0 for x in this equation yields $g0 = 0 - 60 - 20 + 6 + 4$, or $g0 = 76$. Thus, the value of $g0$ is 76.

▼ 3.74 — B

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $gt = a(t - h)^2 + k$, where the maximum value of the function is k, which occurs when $t = h$, and a is a negative constant. The equation $gt = -2t^2 + 55t + 3$ is equivalent to $g(t) = -2(t^2 - \frac{55}{2}t) + 3$, which can be rewritten as $gt = -2(t^2 - \frac{55}{2}t + (\frac{55}{4})^2) + 3 + 2(\frac{55}{4})^2$, or $gt = -2(t - \frac{55}{4})^2 + 3 + 2(\frac{55}{4})^2$. This equation is in the form $gt = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2(\frac{55}{4})^2$. Thus, the maximum value of $g(t)$ is $3 + 2(\frac{55}{4})^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f, which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of

function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f(\frac{55}{4}) = 55(\frac{55}{4}) - 2(\frac{55}{4})^2$, which is equivalent to $f(\frac{55}{4}) = \frac{55^2}{4} - 2(\frac{55^2}{4^2})$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields $f(\frac{55}{4}) = 4(\frac{55^2}{4^2}) - 2(\frac{55^2}{4^2})$, or $f(\frac{55}{4}) = 2(\frac{55^2}{4^2})$, which is equivalent to $f(\frac{55}{4}) = 2(\frac{55}{4})^2$. Thus, the maximum value of $f(t)$ is $2(\frac{55}{4})^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2(\frac{55}{4})^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.75 — A

Choice A is correct. It's given that $f(x) = 272(2)^x$ and $h(x) = f(x - 4)$. Substituting $x - 4$ for x in $f(x) = 272(2)^x$ yields $fx - 4 = 2722^{x-4}$. Substituting hx for $fx - 4$ in this equation yields $hx = 2722^{x-4}$. Using the properties of exponents, the function $h(x) = 272(2)^{x-4}$ can be rewritten as $hx = \frac{2722^x}{2^4}$, which is equivalent to $hx = \frac{2722^x}{16}$, or $h(x) = 17(2)^x$. Therefore, of the given choices, an equation that defines function h is $h(x) = 17(2)^x$.

Choice B is incorrect. This equation defines function h if $h(x) = f(x - 2)$, not $h(x) = f(x - 4)$.

Choice C is incorrect. This equation defines function h if $h(x) = f(4x)$, not $h(x) = f(x - 4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.76 — 5

The correct answer is 5. It's given that the number of bowls sold can be modeled by a quadratic function based on the price, where a maximum of 9 bowls were sold for $\$ 10$ each and 5 bowls were sold for $\$ 12$ each. This quadratic function can be defined by an equation of the form $Np = a(p - h)^2 + k$, where $N(p)$ is the number of bowls sold when the price is $\$ p$ each, and a, h, and k are constants such that the function reaches its maximum value, k, when $p = h$. Since the model indicates the maximum number of bowls sold is 9 when the price is $\$ 10$ each, $N(p)$ reaches its maximum value, 9, when $p = 10$. Therefore, $k = 9$ and $h = 10$. Substituting 10 for h and 9 for k in the function $Np = a(p - h)^2 + k$ yields $Np = a(p - 10)^2 + 9$. Since the model indicates the number of bowls sold is 5 when the price is $\$ 12$ each, the value of $N(p)$ is 5 when $p = 12$. Substituting 12 for p and 5 for $N(p)$ in the equation $Np = a(p - 10)^2 + 9$ yields $5 = a(12 - 10)^2 + 9$, or $5 = 4a + 9$. Subtracting 9 from both

sides of this equation yields $-4 = 4a$. Dividing both sides of this equation by 4 yields $-1 = a$. Therefore, the model can be represented by the equation $Np = -(p - 10)^2 + 9$. Substituting 8 for p in this equation yields $N8 = -(8 - 10)^2 + 9$, or $N(8) = -4 + 9$, or $N(8) = 5$. Therefore, according to the model, the number of bowls sold for $\$8$ each was 5.

▼ 3.77 — C

Choice C is correct. It's given that the function h is defined by $hx = a^x + b$ and that the graph of $y = h(x)$ in the xy-plane passes through the points $0, 10$ and $-2, \frac{325}{36}$. Substituting 0 for x and 10 for hx in the equation $hx = a^x + b$ yields $10 = a^0 + b$, or $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting -2 for x and $\frac{325}{36}$ for hx in the equation $h(x) = a^x + 9$ yields $\frac{325}{36} = a^{-2} + 9$. Subtracting 9 from both sides of this equation yields $\frac{1}{36} = a^{-2}$, which can be rewritten as $a^2 = 36$. Taking the square root of both sides of this equation yields $a = 6$ and $a = -6$, but because it's given that a is a positive constant, a must equal 6. Because the value of a is 6 and the value of b is 9, the value of ab is $(6)(9)$, or 54.

Choice A is incorrect and may result from finding the value of $a^{-2}b$ rather than the value of ab.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from correctly finding the value of a as 6, but multiplying it by the y-value in the first ordered pair rather than by the value of b.

▼ 3.78 — D

Choice D is correct. Each point x, y on the graph of $y = f(x)$ in the xy-plane gives a value of x and its corresponding value of $f(x)$, or y. For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a y-coordinate of 0. A point with a y-coordinate of 0 is a point where the graph intersects the x-axis. It's given that $fx = ab^x + c$, where a, b, and c are constants. In the xy-plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the x-axis. Therefore, there are zero values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 3.79 — B

Choice B is correct. It's given that the situation can be modeled by the equation $y = -100x^2 + 1{,}400x + 7{,}200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy-plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a, b, and c are constants, is graphed in the xy-plane, the x-coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1{,}400x + 7{,}200$, $a = -100$, $b = 1{,}400$, and $c = 7{,}200$. It follows that the x-coordinate of the vertex is $-\frac{1{,}400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy-plane, the maximum of the graph occurs at an x-value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.80 — 7

The correct answer is 7. When an equation is of the form $y = ax^2 + bx + c$, where a, b, and c are constants, the value of y reaches its minimum when $x = -\frac{b}{2a}$. Since the given equation is of the form $y = ax^2 + bx + c$, it follows that $a = 1$, $b = -14$, and $c = 22$. Therefore, the value of y reaches its minimum when $x = -\frac{-14}{21}$, or $x = 7$.

▼ 3.81 — A

Choice A is correct. Let x represent the side length, in inches, of square P. It follows that the perimeter of square P is $4x$ inches. It's given that square Q has a perimeter that is 176 inches greater than the perimeter of square P. Thus, the perimeter of square Q is 176 inches greater than $4x$ inches, or $4x + 176$ inches. Since the perimeter of a square is 4 times the side length of the square, each side length of Q is $\frac{4x + 176}{4}$, or $x + 44$ inches. Since the area of a square is calculated by multiplying the length of two sides, the area of square Q is $x + 44x + 44$, or $x + 44^2$ square inches. It follows that function f is defined by $fx = x + 44^2$.

Choice B is incorrect. This function represents a square with side lengths $x + 176$ inches.

Choice C is incorrect. This function represents a square with side lengths $176x + 44$ inches.

Choice D is incorrect. This function represents a square with side lengths $176x + 176$ inches.

▼ 3.82 — A

Choice A is correct. The graph of a quadratic equation in the form $y = ax - h^2 + k$, where a, h, and k are positive constants, is a parabola that opens upward with vertex h, k. The given function $fx = \frac{1}{9}x - 7^2 + 3$ is in the form $y = ax - h^2 + k$, where $y = fx$, $a = \frac{1}{9}$, $h = 7$, and $k = 3$. Therefore, the graph of $y = fx$ is a parabola that opens upward with vertex $7, 3$. Since the parabola opens upward, the vertex is the lowest point on the graph. It follows that the y-coordinate of the vertex of the graph of $y = fx$ is the minimum value of fx. Therefore, the minimum value of fx is 3. It's given that $fx = \frac{1}{9}x - 7^2 + 3$ represents the metal ball's height above the ground, in inches, x seconds after it started moving on a track. Therefore, the best interpretation of the vertex of the graph of $y = fx$ is that the metal ball's minimum height was 3 inches above the ground.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.83 — 110

The correct answer is 110. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}(2x + 22)(x - 10)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}(2x + 22)(x - 10) = x^2$. This equation can be rewritten as $(x + 11)(x - 10) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding 110 to both sides of this equation yields $x = 110$. Therefore, the value of x is 110.

▼ 3.84 — B

Choice B is correct. For the function f, since the base of the exponent, 1.25, is greater than 1, the value of 1.25^x increases as x increases. Therefore, the value of 181.25^x and the value of $181.25^x + 41$ also increase as x increases. Since f is therefore an increasing function where $x \geq 0$, the function f has no maximum value. For the function g, since the base of the exponent, 0.73, is less than 1, the value of 0.73^x decreases as x increases. Therefore, the value of 90.73^x also decreases as x increases. It follows that the maximum value of gx for

$x \geq 0$ occurs when $x = 0$. Substituting 0 for x in the function g yields $g0 = 90.73^0$, which is equivalent to $g0 = 91$, or $g0 = 9$. Therefore, the maximum value of gx for $x \geq 0$ is 9, which appears as a coefficient in equation II. So, of the two equations given, only II displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.85 — B

Choice B is correct. It's given that $y = 9\frac{a}{7}^{x+c} - b$, where a, b, and c are positive constants such that $a > 7$ and $b > c$. The graph of any equation in the xy-plane crosses the x-axis when there is a point on the graph for which the value of y is equal to 0, and the value of y changes sign when the graph passes through that point. Since $a > 7$, the base $\frac{a}{7}$ is greater than 1, so the expression $9\frac{a}{7}^{x+c} - b$ increases as x increases. Since b is positive, there is a point on the graph of $y = 9\frac{a}{7}^{x+c} - b$ for which the value of y is equal to 0. When the graph passes through this point, the value of y changes sign from negative to positive. Therefore, the graph of the given equation crosses the x-axis exactly one time.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3.86 — 35

The correct answer is 35. It's given that the function $gx = -\frac{1}{55}x + 19x - 35$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. It follows that when the submersible device is at the surface of the ocean, the value of gx is 0. Substituting 0 for gx in the equation $gx = -\frac{1}{55}x + 19x - 35$ yields $0 = -\frac{1}{55}x + 19x - 35$. Multiplying both sides of this equation by -55 yields $0 = x + 19x - 35$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $x + 19 = 0$ or $x - 35 = 0$. Subtracting 19 from both sides of the equation $x + 19 = 0$ yields $x = -19$. Adding 35 to both sides of the equation $x - 35 = 0$ yields $x = 35$. Since $x > 0$, 35 minutes after collecting the sample the submersible device reached the surface of the ocean.